

```
D:\MyDrive\Learning\Repo\requirement_analysis_agent>python RAG_req_analysis_agent.py -f requirements.txt -k Core_v6.0.pdf
Requirements Ambiguity Analysis Agent
```

```
Building knowledge base from 1 file(s)...
```

Loading embedding model all-MiniLM-L6-v2...

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
```

```
Loading weights: 100% | 103/103 [00:00<00:00, 5354.59it/s]
```

Reading Core_v6.0.pdf

→ 19752 chunks

Embedding 19752 chunks...

[illegible]

```
D:\MyDrive\Learning\Repo\requirement_analysis_agent>python main.py -f .\others\requirements.txt -k Core_v6.0.pdf
```

Building knowledge base - 1 file(s) found

```
Loading embedding model all-MiniLM-L6-v2 ...
```

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.

[illegible]

Reading Core_v6.0.pdf

→ 19752 chunks

Embedding 19752 chunks ...

[illegible]

```
Storing chunks in ChromaDB ...
```

Batch 1/4 - chunks 1 to 5000 stored

Batch 2/4 - chunks 5001 to 10000 stored

Batch 3/4 - chunks 10001 to 15000 stored

Batch 4/4 - chunks 15001 to 19752 stored

✓ Knowledge base ready - 19752 chunks indexed

```
Requirements from: others\requirements.txt
```

Analysing 4 requirement(s) – one at a time

```
→ REQ-01: The system shall support Apple CarPlay and Android Auto via both wired...
JSON parse error on REQ-01: Expecting property name enclosed in double quotes: line 12 column 42 (char 502)
→ REQ-02: Bluetooth 5.0 or higher shall be supported for hands-free calling and ...
→ REQ-03: The system shall support simultaneous pairing of up to 5 Bluetooth dev...
→ REQ-04: Wi-Fi hotspot capability (4G/5G) shall be available as a built-in or t...
```

Analysis Summary

Total requirements analysed	3
Need more info (High + Medium)	1
High ambiguity	1
Medium ambiguity	0
Low ambiguity	2
Clear & testable	0

Bluetooth 5.0 or higher shall be supported for hands-free calling and audio streaming.

Knowledge base: Core_v6.0.pdf

- What is missing:

- Vague terms:

Missing acceptance criteria:

- No RSSI threshold defined for 'hands-free calling' - typical BLE proximity is -70 dBm to -90 dBm (NIST SP 800-63B)

⚠ Edge cases not covered:

? Ask the product owner:

Which specific Bluetooth version range (e.g., 5.0, 5.1, 5.2, 5.3) must be supported?

Are there any specific audio streaming requirements or codecs that must be supported?

● REQ-03 [High]

The system shall support simultaneous pairing of up to 5 Bluetooth devices with automatic reconnection to the last active device.

Knowledge base: Core_v6.0.pdf

What is missing:

- No definition of 'last active device' – how is it determined?
- No explanation for what constitutes 'automatic reconnection'
- No threshold for maximum number of simultaneous pairings (5) – why not 4 or more?
- No consideration for devices with limited power resources

Vague terms:

- "last active device" – Lack of clarity on how to determine the last active device
 - ↳ ['Device with most recent pairing attempt', 'Device that was paired first']
- "automatic reconnection" – Vagueness in definition of automatic reconnection process
 - ↳ ['Reconnect after 10 seconds of inactivity', 'Reconnect immediately when device is lost from range']

Missing acceptance criteria:

- No RSSI threshold defined for simultaneous pairing – typical BLE proximity is -70 dBm to -90 dBm

Edge cases not covered:

- Behaviour when Bluetooth is disabled on the device
- What happens when two devices attempt to pair simultaneously with more than 5 devices already paired
- Behaviour at maximum connection range boundary

? Ask the product owner:

How does the system determine which device to reconnect to if multiple devices are active?
What is the maximum allowed time for automatic reconnection?

● REQ-04 [Low]

Wi-Fi hotspot capability (4G/5G) shall be available as a built-in or tethered option.

Knowledge base: Core_v6.0.pdf

What is missing:

- No specific frequency band specified for Wi-Fi hotspot (2.4GHz, 5GHz, etc.)

Vague terms:

- "Wi-Fi hotspot capability" – Lack of specificity on supported frequency bands and network protocols
 - ↳ ['Supporting only 2.4GHz', 'Supporting both 2.4GHz and 5GHz']

Missing acceptance criteria:

- No definition of acceptable signal strength for Wi-Fi hotspot connection

Edge cases not covered:

- Behaviour when Bluetooth is disabled on the device
- What happens when two devices attempt to pair simultaneously while using Wi-Fi hotspot

? Ask the product owner:

Which frequency bands (2.4GHz, 5GHz) must be supported for Wi-Fi hotspot capability?
How will the device determine acceptable signal strength for Wi-Fi hotspot connection?

Suggested rewrite:

Wi-Fi hotspot capability (4G/5G) shall be available as a built-in option supporting both 2.4GHz and 5GHz frequency bands, with a minimum signal strength of -70 dBm for acceptable connection.

Batches: 100% | Traceback (most recent call last) | 618/618 [10:07<00:00, 1.02it/s]

D:\MyDrive\Learning\Repo\requirement_analysis_agent\RAG_req_analysis_agent.py:489 in main

```

486
487     # Build knowledge base if provided
488     if knowledge:
489         has_kb = build_knowledge_base(list(knowledge))
490         if not has_kb:
491             console.print("[yellow]Continuing without knowledge base.[/yellow]\n")
492     else:

```

Locals

```

file = WindowsPath('requirements.txt')
interactive = False
knowledge = [WindowsPath('Core_v6.0.pdf')]
model = 'llama3.2'
output = None
top_k = 5

```

D:\MyDrive\Learning\Repo\requirement_analysis_agent\RAG_req_analysis_agent.py:171 in build_knowledge_base

```

168     # Build in-memory ChromaDB collection (no disk, no server)
169     chroma = chromadb.Client()
170     _collection = chroma.create_collection("knowledge_base")
171     _collection.add(ids=ids, documents=texts, embeddings=embeddings, metadatas=metadatas)
172
173     console.print(f"\n [green]✓ Knowledge base ready[/green] - {len(all_chunks)} chunks
174     return True

```

```

locals
all_chunks = [
    {
        'id': 'Core_v6.0.pdf::chunk_0',
        'text': 'Bluetooth Core Specification Bluetooth® Specification ▪ Version: v6.0 ▪ Version '+320,
        'source': 'Core_v6.0.pdf'
    },
    {
        'id': 'Core_v6.0.pdf::chunk_1',
        'text': 'this specification is your acknowledgement that you agree to and will comply wit'+319,
        'source': 'Core_v6.0.pdf'
    },
    {
        'id': 'Core_v6.0.pdf::chunk_2',
        'text': 'of Bluetooth SIG is governed by the membership and other related agreements betw'+320,
        'source': 'Core_v6.0.pdf'
    },
    {
        'id': 'Core_v6.0.pdf::chunk_3',
        'text': 'ed agreements is prohibited and, among other things, may result in (i) terminati'+320,
        'source': 'Core_v6.0.pdf'
    },
    {
        'id': 'Core_v6.0.pdf::chunk_4',
        'text': 'tion of the specification. All content within the specification, including notes'+320,
        'source': 'Core_v6.0.pdf'
    },
    {
        'id': 'Core_v6.0.pdf::chunk_5',

```

```

        'id': 'Core_v6.0.pdf::chunk_9',
        'text': 'DISCLAIM ALL LIABILITY ARISING OUT OF OR RELATING TO USE OF THIS SPECIFICATION A'+319,
        'source': 'Core_v6.0.pdf'
    },
    ... +19742
]
chroma = <chromadb.api.client.Client object at 0x000002B1BEC55010>
chromadb = <module 'chromadb' from 'D:\\Python_Installation\\Lib\\site-packages\\chromadb\\__init__.py'>
chunks = [
    {
        'id': 'Core_v6.0.pdf::chunk_0',
        'text': 'Bluetooth Core Specification Bluetooth® Specification ▪ Version: v6.0 ▪ Version '+320,
        'source': 'Core_v6.0.pdf'
    },
    {
        'id': 'Core_v6.0.pdf::chunk_1',
        'text': 'this specification is your acknowledgement that you agree to and will comply wit'+319,
        'source': 'Core_v6.0.pdf'
    },
    {
        'id': 'Core_v6.0.pdf::chunk_2',

```

```

    },
    {
        'id': 'Core_v6.0.pdf::chunk_9',
        'text': 'DISCLAIM ALL LIABILITY ARISING OUT OF OR RELATING TO USE OF THIS SPECIFICATION A'+319,
        'source': 'Core_v6.0.pdf'
    },
    ... +19742
]
embedder = SentenceTransformer(
    (0): Transformer({'transformer_task': 'feature-extraction', 'modality_config': {'text': {'method': 'forward',
'method_output_name': 'last_hidden_state'}}}, 'module_output_name': 'token_embeddings', 'architecture':
'BertModel'})
    (1): Pooling({'embedding_dimension': 384, 'pooling_mode': 'mean', 'include_prompt': True})
    (2): Normalize({})
)
embeddings = [
    [
        -0.06014426797628403,
        -0.05457574129104614,
        0.057117071002721786,
        -0.06752721965312958,
        0.009270467795431614,
        0.02024068497121334,
        -0.005893660243600607,
        0.06428588926792145,
        -0.08577819913625717,
        -0.018766460940241814,
        ... +374
    ]

```

```

    ],
    [
        -0.06053297966718674,
        0.06461784988641739,
        0.03927989304065704,
        -0.06300479173660278,
        0.04237858206033707,
        0.060233402997255325,
        0.030310282483696938,
        0.09683424234390259,
        -0.08593232184648514,
        0.04898436367511749,
        ... +374
    ],
    ... +19742
]
f = WindowsPath('Core_v6.0.pdf')
files = [WindowsPath('Core_v6.0.pdf')]
ids = [
    'Core_v6.0.pdf::chunk_0',
    'Core_v6.0.pdf::chunk_1',
    'Core_v6.0.pdf::chunk_2',
    'Core_v6.0.pdf::chunk_3',
    'Core_v6.0.pdf::chunk_4',
    'Core_v6.0.pdf::chunk_5',
    'Core_v6.0.pdf::chunk_6',
    'Core_v6.0.pdf::chunk_7',
    'Core_v6.0.pdf::chunk_8',
    'Core_v6.0.pdf::chunk_9',
    ... +19742
]

```

```

    ... +19742
]
kb_paths = [WindowsPath('Core_v6.0.pdf')]
metadatas = [
    {'source': 'Core_v6.0.pdf'},
    {'source': 'Core_v6.0.pdf'},
    {'source': 'Core_v6.0.pdf'},
    {'source': 'Core_v6.0.pdf'},
    {'source': 'Core_v6.0.pdf'},
    {'source': 'Core_v6.0.pdf'},
    {'source': 'Core_v6.0.pdf'},
    {'source': 'Core_v6.0.pdf'},
    {'source': 'Core_v6.0.pdf'},
    {'source': 'Core_v6.0.pdf'},
    ... +19742
]
text = 'Bluetooth Core Specification\nBluetooth® Specification\n Version: v6.0\n Version '+6417167
texts = [
    'Bluetooth Core Specification Bluetooth® Specification ▪ Version: v6.0 ▪ Version '+320,
    'this specification is your acknowledgement that you agree to and will comply wit'+319,
    'of Bluetooth SIG is governed by the membership and other related agreements betw'+320,
    'ed agreements is prohibited and, among other things, may result in (i) terminati'+320,
    'tion of the specification. All content within the specification, including notes'+320,
    'entification of options for implementing a portion of the specification is inten'+320,
    's prohibited and is an infringement of the intellectual property rights of Bluet'+320,
    'ES MAKE NO REPRESENTATIONS OR WARRANTIES AND DISCLAIM ALL WARRANTIES, EXPRESS OR'+320,
    'ch or investigation as to third parties that may claim rights in or to any speci'+319,
    'DISCLAIM ALL LIABILITY ARISING OUT OF OR RELATING TO USE OF THIS SPECIFICATION A'+319,
    ... +19742
]

```

```
'ch or investigation as to third parties that may claim rights in or to any speci'+319,  
'DISCLAIM ALL LIABILITY ARISING OUT OF OR RELATING TO USE OF THIS SPECIFICATION A'+319,  
... +19742
```

D:\Python_Installation\Lib\site-packages\chromadb\api\models\Collection.py:118 in add

```
115 |         uris=uris,  
116 |     )  
117 |  
> 118 |     self._client._add(  
119 |         collection_id=self.id,  
120 |         ids=add_request["ids"],  
121 |         embeddings=add_request["embeddings"],
```

locals

```
add_request = {  
    'ids': [  
        'Core_v6.0.pdf::chunk_0',  
        'Core_v6.0.pdf::chunk_1',  
        'Core_v6.0.pdf::chunk_2',  
        'Core_v6.0.pdf::chunk_3',  
        'Core_v6.0.pdf::chunk_4',  
        'Core_v6.0.pdf::chunk_5',  
        'Core_v6.0.pdf::chunk_6',  
        'Core_v6.0.pdf::chunk_7',  
        'Core_v6.0.pdf::chunk_8',  
        'Core_v6.0.pdf::chunk_9',  
        ... +19742  
    ],  
    'embeddings': [  
        array([-6.01442680e-02, -5.45757413e-02,  5.71170710e-02, -6.75272197e-02,  
            9.27046780e-03,  2.02406850e-02, -5.89366024e-03,  6.42858893e-02,  
           -8.57781991e-02, -1.87664609e-02,  4.35518064e-02, -2.33548284e-02,  
            3.89540419e-02, -4.39394340e-02,  6.25820532e-02, -8.23462557e-04,  
            4.77527604e-02, -8.55001360e-02, -2.49610376e-02, -2.11567283e-02,  
            3.70730124e-02,  3.15484777e-02, -1.73753798e-02,  5.21856174e-02,  
           -9.52734204e-04, -2.22959351e-02,  4.21170145e-02, -1.94489155e-02,  
           -2.25292929e-02,  7.76112229e-02, -5.50241508e-02,  4.35861349e-02,  
            1.10238396e-01,  2.70309560e-02, -2.73838770e-02, -8.11522454e-02,  
            2.24103630e-02, -8.16603452e-02, -9.63936299e-02, -5.59672974e-02,  
           -8.39130357e-02, -7.92955384e-02, -3.57170589e-02,  3.28305364e-02,  
            7.95071071e-04,  4.79242802e-02, -6.68670610e-02, -8.93762801e-03,  
           -4.17377725e-02,  8.65854695e-02,  9.45833102e-02, -6.15602285e-02,  
            4.89731655e-02, -2.33542155e-02, -6.44337684e-02,  4.25465368e-02,  
            5.75682754e-03, -4.78228927e-02, -1.87631585e-02, -2.85343342e-02,  
           -5.72205931e-02,  4.34369259e-02, -3.18089202e-02, -3.35709527e-02,  
            5.88976927e-02,  2.27949372e-03,  6.71485290e-02,  5.80243692e-02,  
            7.71894902e-02, -6.49899021e-02,  1.29888922e-01, -3.22737806e-02,  
            3.49379256e-02,  4.73490432e-02,  5.97718023e-02,  8.42149109e-02],  
        dtype=float32),  
        array([-6.36326447e-02,  3.29842865e-02,  7.98410624e-02, -5.95186763e-02,  
           -1.24166496e-02,  6.15250431e-02,  7.09862486e-02,  8.06638971e-02,  
           -8.55943188e-02, -4.94908951e-02,  6.04855418e-02, -3.56492437e-02,  
            7.14632720e-02, -3.18725966e-02,  4.26005125e-02, -1.68751515e-02,  
            5.51460534e-02, -5.73281422e-02, -8.88746828e-02, -4.60871076e-03,  
            4.55264151e-02,  6.35164306e-02, -2.03312226e-02,  1.07245013e-01,  
           -7.00924918e-02, -5.00290133e-02,  5.52615412e-02,  4.71812449e-02,  
            1.73684806e-02,  5.43966666e-02, -5.52360490e-02,  5.85107282e-02,  
            1.34585977e-01, -2.53873933e-02, -4.69813645e-02, -6.54518828e-02,  
            4.68331343e-03, -9.02610272e-02, -9.90822315e-02, -9.28296894e-03,  
           -7.04926029e-02, -8.19861740e-02, -5.12899458e-02,  6.82712421e-02,
```

```

    7.14969262e-02, -2.12307870e-02, 2.18634643e-02, -4.85277958e-02,
    4.82851686e-03, -2.63077151e-02, 5.98335937e-02, 6.35535782e-03],
    dtype=float32),
    ... +19742
]
ids = [
    'Core_v6.0.pdf::chunk_0',
    'Core_v6.0.pdf::chunk_1',
    'Core_v6.0.pdf::chunk_2',
    'Core_v6.0.pdf::chunk_3',
    'Core_v6.0.pdf::chunk_4',
    'Core_v6.0.pdf::chunk_5',
    'Core_v6.0.pdf::chunk_6',
    'Core_v6.0.pdf::chunk_7',
    'Core_v6.0.pdf::chunk_8',
    'Core_v6.0.pdf::chunk_9',
    ... +19742
]
metadatas = [
    {'source': 'Core_v6.0.pdf'},
    {'source': 'Core_v6.0.pdf'},
    {'source': 'Core_v6.0.pdf'},
    {'source': 'Core_v6.0.pdf'},
    {'source': 'Core_v6.0.pdf'},
    {'source': 'Core_v6.0.pdf'},
    {'source': 'Core_v6.0.pdf'},
    {'source': 'Core_v6.0.pdf'},
    {'source': 'Core_v6.0.pdf'},
    {'source': 'Core_v6.0.pdf'},
    ... +19742
]
self = <chromadb.api.rust.RustBindingsAPI object at 0x000002B1BEC557F0>
tenant = 'default_tenant'
uris = None

```

InternalError: ValueError: Batch size of 19752 is greater than max batch size of 5461